HyQ-LC: Improving Light-Curve Classification with Hybrid Quantum–Classical Learning

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Light curves reveal stellar variability and support studies of stellar evolution and cosmic distances. Reliable classification is therefore important for large time-domain surveys. We present HyQ-LC for ten-class stellar light-curve classification using 359 handcrafted features and 633,034 training objects. The main model combines a window multilayer perceptron (MLP) backbone with a nine-qubit readout using amplitude encoding and angle re-uploading. Both this hybrid and its pure-MLP control have 44,238 parameters. On a shared 20,000-object test set, the hybrid improves macro-averaged F1 (Macro-F1) by 0.8276 and 1.2393 percentage points on two fine-tuning seeds. A separate fixed-fusion configuration achieves slightly higher Macro-F1 but lower accuracy than approximately budget-matched LightGBM. Its positive class-level differences are concentrated in CEP, RRC, and ROT, while the full-budget LightGBM reference scores higher on both metrics. All circuits are simulated classically. The comparisons retain existing validation-selected checkpoints and use the same test objects, with features, predictions, and recomputation code publicly available. The results identify improvements over the tested controls while leaving the specific contribution of the quantum branch unresolved.

1 Introduction

Time-domain surveys generate large collections of light curves that require scalable classification [1]. Feature-based systems such as ALeRCE [2] and incremental frameworks such as TDLight [3] address this task through complementary classification and data-management strategies. Learned representations provide another route, including ASTROMER embeddings [4], time-series foundation models benchmarked by StarEmbed [5], and structured sequence modelling with large language models [6]. These studies provide context; their different inputs, labels, and evaluation protocols prevent direct score comparisons with the present task.

Hybrid quantum–classical learning combines classical processing with trainable quantum representations [7]. Peters et al. demonstrated quantum-kernel classification of supernova spectral features on a noisy processor [8]. Architecture comparisons on tabular data also motivate studying how quantum branches interact with classical networks [9]. However, expressive circuits alone do not guarantee predictive improvements. Bowles et

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al. showed that classical baselines and entanglement-removal controls can substantially change the interpretation of quantum-learning benchmarks [10].

Here we ask whether a hybrid representation improves stellar light-curve classification under controlled parameter budgets. We distinguish comparisons with a parameter-matched pure MLP from replacements of the quantum branch within a shared network. A separate fixed-fusion configuration supports comparisons with classical fusion and an approximately budget-matched LightGBM. Every reported test comparison uses the same 20,000 objects.

2 Models and Evaluation

2.1 Data and test cohort

The task contains nine variable-star categories and one non-variable category. The labels are CEP, DSCT, EA, EW, M, Non-var, ROT, RRAB, RRC, and SR; Non-var denotes non-variable objects. Each object has 359 precomputed handcrafted features. The training pool contains 633,034 objects, and the reported test cohort contains 20,000 unique objects. EW and SR together account for 61.89% of the test cohort; CEP has only 45 objects. Complete class counts are provided with the released data [11].

Preprocessing uses frozen training-derived medians and scales, median replacement of non-finite values, and clipping to $[-10, 10]$. The main models use 105,297 validation objects; the fixed-fusion and leaf-budget LightGBM experiments use 10,000. We retain their existing validation-selected checkpoints and hyperparameters. Test predictions are extracted by exact ID and label matching from saved prediction files. The cohort was previously examined during exploratory development, so it is not an untouched confirmation sample. Separate sealed confirmation data remain unused.

2.2 Residual hybrid and parameter-matched controls

The main HyQ-LC model has 44,238 parameters (Figure 1a). Its classical backbone is a 23,170-parameter window MLP. After an elementwise arctan transform, the features are padded into 120 three-component tokens. Token MLPs and two stages of six-component window mixing produce a 120-dimensional representation and ten base logits.

The quantum branch also receives the transformed input. It pads the 359 values to 512 entries and normalises them as amplitudes of a nine-qubit state. Group means define nine additional input angles. Each of twelve layers applies trainable single-qubit rotations, a ring of CNOT gates, and angle re-uploading [12]. The readout contains 324 rotation parameters and 216 re-uploading frequencies and shifts. Measuring single-qubit X, Y, and Z expectations produces 27 features [13]. Layer-normalised classical and quantum features enter a $147 \to 128 \to 10$ MLP whose output is added to the backbone logits. Encoding affects the accessible function class [14], but depth and expressivity do not by themselves establish predictive performance [15].

The pure-MLP control has the same 44,238 parameters, with hidden widths 73 and 207, LayerNorm, and GELU activations. It is trained from scratch under an 80-epoch budget. The hybrid uses 40 backbone pre-training epochs followed by a 40-epoch branch-training stage. The backbone is frozen for the first ten epochs of that stage; the primary selected hybrid checkpoints occur at epochs two and four. Thus, the budgets match the maximum number of training-object presentations, not training trajectories or computational cost. The two primary fine-tuning seeds, 42 and 44, share the pre-trained backbone but vary added weights and sample order.

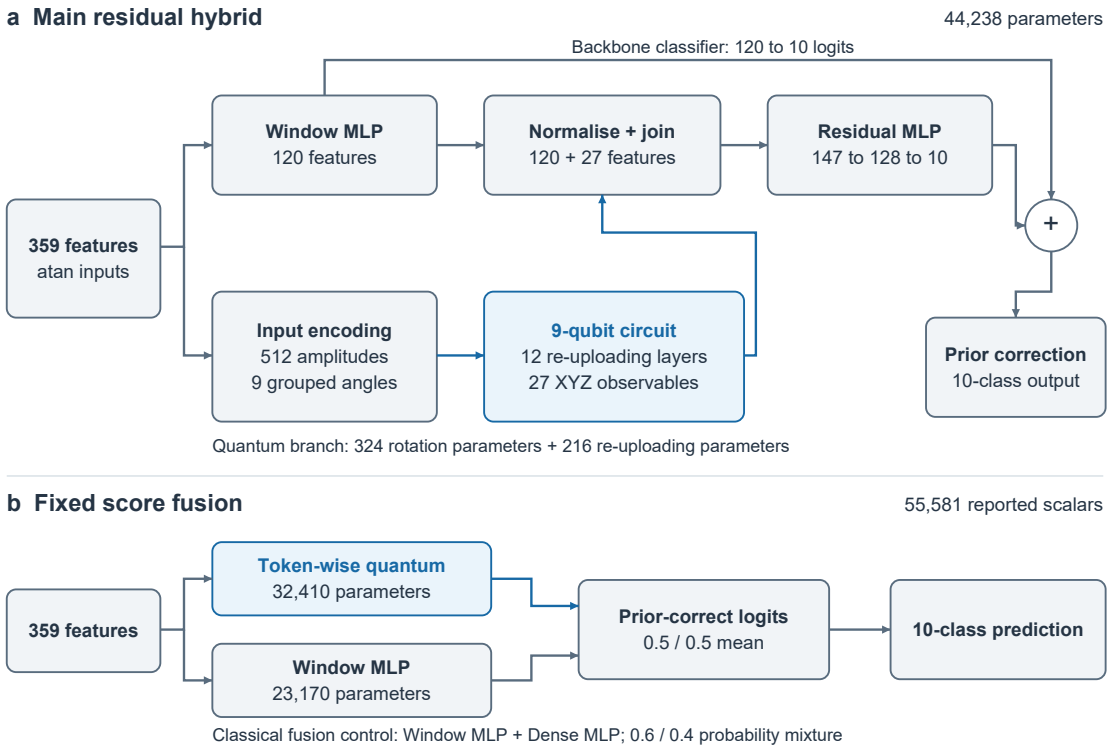


Figure 1: Two distinct hybrid configurations. (a) The main model adds a residual prediction from classical features and a global nine-qubit readout to the backbone logits. The quantum input combines normalised amplitudes and grouped angles. (b) Fixed fusion averages the corrected logits of two independently trained classifiers. Its classical control instead mixes Window MLP and Dense MLP probabilities. The fixed-fusion scalar budgets include one fixed mixing coefficient.

For the branch control, a 540-parameter MLP replaces the quantum readout. It receives a fixed 26-dimensional projection of the normalised input and grouped angles. The backbone, normalisation layers, fusion-head initialisation, parameter total, optimiser-group allocation, and sample order are preserved. This comparison tests the two branch constructions within a shared fusion network.

2.3 Fixed fusion and LightGBM controls

The fixed-fusion configuration combines a 32,410-parameter token-wise quantum classifier with the 23,170-parameter window MLP (Figure 1b). Their corrected logits are averaged with equal weights. The classical control combines the window MLP with a 31,460-parameter Dense MLP, using a 0.6/0.4 probability mixture. Both fusion rules were selected on validation. Reported budgets are 55,581 and 54,631 scalars, including one fixed mixing coefficient each. Because the fusion rules and parameter totals differ, this is an approximate-budget ensemble comparison, rather than an isolated branch replacement. Each constituent network receives 40 training epochs.

The LightGBM [16] control targets the fixed hybrid's scalar budget using cumulative leaf outputs across all class trees. The selected model has 15 leaves and 370 boosting rounds, giving 3,700 class trees and 55,500 leaf values. This proxy excludes split-node features and thresholds and does not equate tree capacity with neural parameters or runtime. A separate full-budget reference has 31 leaves and 362 rounds, with 112,220 leaf values. Both references retain their validation-selected settings.

2.4 Selection, metrics, and uncertainty

Neural models use inverse-frequency class-weighted cross-entropy. Before prediction, class logits are corrected as $z_c - \alpha \log w_c$, where $w_c = N/(10n_c)$ uses training-class counts. Checkpoints and $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ are selected by validation Macro-F1. For each pure-MLP seed, the learning rate is selected between 10^{-3} and 3×10^{-3} using validation only. ACC in the manuscript tables uses the same Macro-F1-selected checkpoint. Circuit evaluations use classical CPU/GPU state-vector simulation, with no quantum hardware.

Macro-F1 gives equal weight to the ten classwise F1 scores; accuracy (ACC) is the fraction of correctly classified objects. Differences are hybrid minus control in percentage points (pp). We use 10,000 ordinary paired-object bootstrap resamples and percentile 95% confidence intervals. Each resample uses the same objects for both models, with class proportions allowed to fluctuate. The intervals are conditional on the fitted models and exclude training-seed and model-search variation; no multiplicity correction is applied. The release also includes additional seeded runs and numerical ablations.

3 Results

3.1 Comparison with the parameter-matched pure MLP

The residual hybrid had higher Macro-F1 and accuracy than the pure MLP on both primary seeds (Table 1). The F1 interval included zero for seed 42 and was positive for seed 44; both ACC intervals were positive. These differences compare the complete trained configurations, including their different architectures and pre-training histories.

Table 1: Parameter-matched comparison on 20,000 test objects. Both models have 44,238 parameters. F1 scores are percentages; differences are in pp and use unrounded scores. Brackets give paired-bootstrap 95% confidence intervals.

Seed	Hybrid F1	MLP F1	ΔF1 (pp)	F1 95% interval	ΔACC (pp), 95% interval
42	92.2913	91.4637	+0.8276	[−0.0650, +1.6945]	+0.6000 [+0.3450, +0.8550]
44	92.3005	91.0611	+1.2393	[+0.5153, +2.0053]	+0.8450 [+0.5900, +1.1000]

3.2 Replacing the quantum branch

Within the shared residual architecture, the quantum readout had slightly higher F1 point estimates than the matched classical readout (Table 2). Both intervals included zero. This test therefore leaves the predictive contribution specific to the quantum branch unresolved.

Table 2: Readout replacement within the residual hybrid on the same test objects. Both readouts have 540 parameters and both full models have 44,238 parameters. Scores are Macro-F1 (%); differences and intervals are in pp.

Seed	Quantum readout	MLP readout	ΔF1 (pp)	95% interval
42	92.2913	92.2708	+0.0205	[−0.3299, +0.3726]
44	92.3005	92.0039	+0.2965	[−0.0333, +0.6982]

3.3 Fixed fusion and tree controls

The fixed quantum fusion achieved 92.3371% Macro-F1 and 94.7450% ACC (Table 3). It had a small F1 point gain over classical fusion, with essentially unchanged ACC. Against leaf-budget-matched LightGBM, its F1 was 0.1166 pp higher but ACC was 0.1200 pp lower, corresponding to 24 fewer correct classifications. The paired F1 intervals spanned zero for both controls. The full-budget LightGBM reference scored higher on both metrics.

The largest positive class-level F1 differences against the budget-matched tree were in CEP (+1.0040 pp), RRC (+1.1429 pp), and ROT (+0.5128 pp). CEP and RRC contain only 45 and 357 test objects. Equal class weighting allows these gains to raise Macro-F1 even when total correct predictions decrease. This is a class-specific pattern, not an improvement across all minority classes.

Table 3: Controls for fixed quantum fusion on the same 20,000 test objects. The hybrid reference scores F1/ACC of 92.3371/94.7450%. Control scores are percentages; hybrid-minus-control differences and paired 95% intervals are in pp.

Control	F1/ACC (%)	ΔF1 (pp), 95% interval	ΔACC (pp)
Classical fusion	92.1510 / 94.7500	$+0.1860$ $[-0.2869, +0.6740]$	-0.0050
Budget-matched LightGBM	92.2204 / 94.8650	$+0.1166$ $[-0.8111, +1.0763]$	-0.1200
Full-budget LightGBM	92.9681 / 95.1100	-0.6311 $[-1.4192, +0.1197]$	-0.3650

4 Discussion and Conclusion

HyQ-LC improved the tested parameter-matched pure-MLP comparison, while branch replacement did not resolve a benefit specific to the quantum readout. The fixed-fusion results show a trade-off between class-balanced F1 and accuracy; the full-budget tree scored higher on both metrics. Parameter counts alone therefore provide an incomplete account of performance: architecture, pre-training, fusion rules, and control strength all affect the comparison. Evaluation on previously unexamined objects and independently pre-trained backbones would test how broadly these observations persist. The present simulations establish neither a hardware speed advantage nor a general quantum advantage.

5 Data and Code Availability

The repository provides the shared test features, labels, per-object predictions, result tables, and a portable metric-recomputation script [11]. Checkpoint and source-prediction hashes document provenance. Standalone quantum-readout code and original training implementation references are included; the latter require the authors' cluster environment. The release supports independent recomputation of the reported statistics from saved predictions.

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